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***Abstract*—**

***Index Terms*—Deep Learning, Dengue Fever, Artificial Neural Network, Classification, Normalization, Cross Validation**

I. Introduction

Dengue Fever is one of the major infectious diseases in tropical countries and remains a serious public health concern. Early classification of Dengue Fever is essential to reduce mortality and improve patient treatment. Conventional diagnosis generally relies on clinical symptoms and laboratory examinations, which may require additional time and expert interpretation.

Recent developments in Artificial Intelligence and Deep Learning have shown promising performance in healthcare classification systems. Artificial Neural Network (ANN) models are capable of learning complex non-linear relationships within medical datasets.

Several studies have applied Machine Learning approaches for Dengue Fever prediction. However, limited studies comprehensively analyze the influence of normalization techniques and cross-validation scenarios on Deep Learning performance.

Therefore, this study proposes a comparative analysis of normalization techniques and cross-validation scenarios for Deep Learning-based Dengue Fever classification using clinical symptom and laboratory datasets.

A. Research Contributions

The contributions of this study are as follows:

- Comparative analysis of normalization techniques for Dengue Fever classification.
- Comparative analysis between full dataset and symptom-only dataset.
- Evaluation of ANN model stability using K-Fold Cross Validation.
- Development of Deep Learning classification model for healthcare decision support systems.

II. Related Work

III. Research Methodology

A. Dataset

The dataset used in this study consists of:

- 9 symptom features
- 4 laboratory features

TABLE I
SUMMARY OF RELATED STUDIES

Study	Method	Accuracy
Study A	ANN	92%
Study B	CNN	94%
Study C	Random Forest	89%

TABLE II
DATASET FEATURES

Category	Number of Features
Symptoms	9
Laboratory	4

B. Dataset Scenarios

C. Data Preprocessing

The preprocessing stage includes:

- Data cleaning
- Missing value handling
- Feature encoding
- Data normalization

D. Normalization Experiments

Three normalization methods were evaluated:

- Without Normalization
- StandardScaler
- MinMaxScaler

E. Proposed ANN Model

F. Cross Validation

K-Fold Cross Validation with K=5 was implemented to evaluate model stability and generalization capability.

TABLE III
DATASET SCENARIOS

Scenario	Dataset Composition
Scenario 1	Symptoms + Laboratory Features
Scenario 2	Symptoms Only

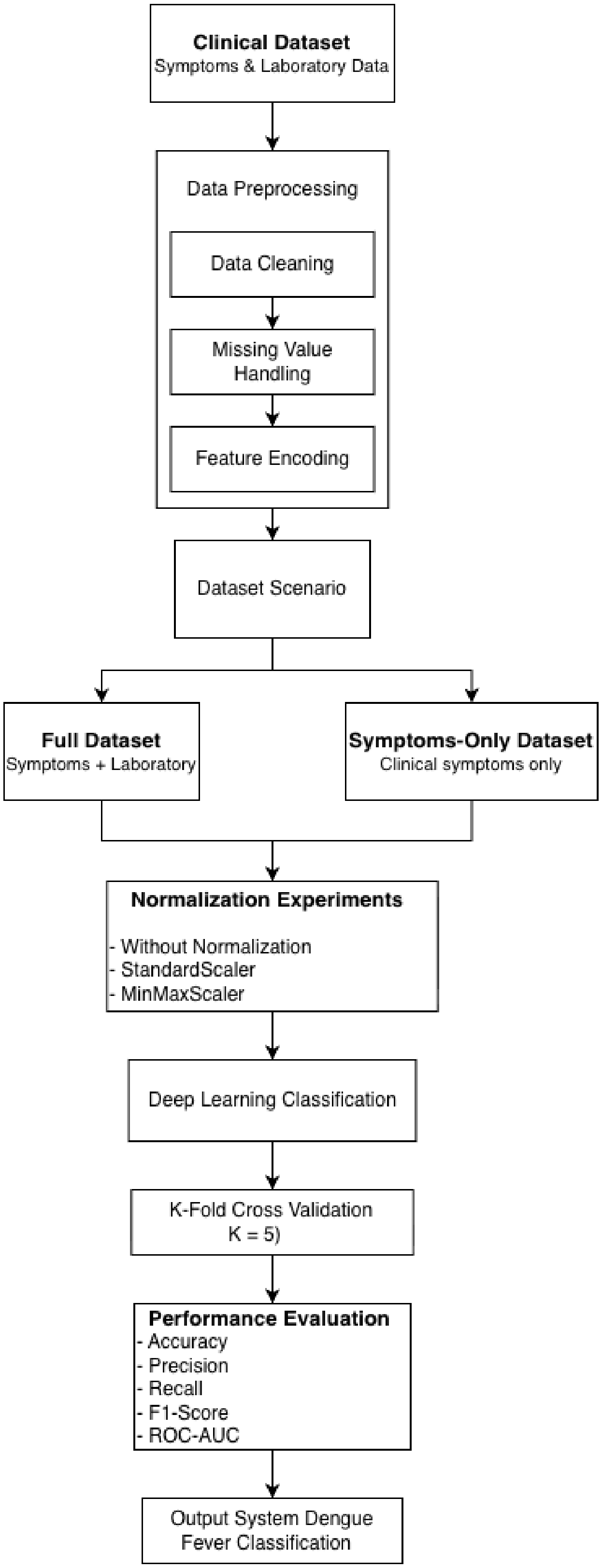

Fig. 1. Research Workflow

TABLE IV
NORMALIZATION EXPERIMENT SCENARIOS

Experiment	Dataset	Normalization
Exp-1	Full Dataset	None
Exp-2	Full Dataset	StandardScaler
Exp-3	Full Dataset	MinMaxScaler
Exp-4	Symptoms Only	None
Exp-5	Symptoms Only	StandardScaler
Exp-6	Symptoms Only	MinMaxScaler

TABLE V
ANN ARCHITECTURE

Layer	Neurons	Activation
Input	13	-
Dense 1	64	ReLU
Dropout	-	-
Dense 2	32	ReLU
Output	1	Sigmoid

G. Evaluation Metrics

The evaluation metrics used in this study include:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

IV. RESULTS AND DISCUSSION

A. Training Performance

The training results demonstrate stable convergence between training and validation processes. The model successfully minimized classification error during training.

B. Normalization Experiment Results

The experimental results indicate that normalization significantly affects ANN classification performance. The Full Dataset scenario consistently achieved better performance than the Symptoms-Only scenario because laboratory features provide additional clinical information.

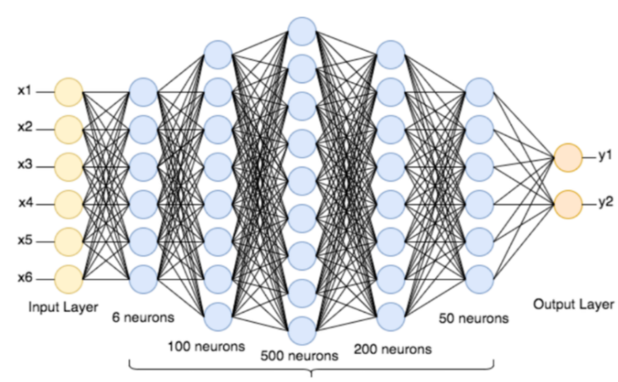

Fig. 2. Artificial Neural Network Architecture

TABLE VI
CROSS VALIDATION SCENARIOS

Scenario	Dataset	Normalization
CV-1	Full Dataset	None
CV-2	Full Dataset	StandardScaler
CV-3	Full Dataset	MinMaxScaler
CV-4	Symptoms Only	None
CV-5	Symptoms Only	StandardScaler
CV-6	Symptoms Only	MinMaxScaler

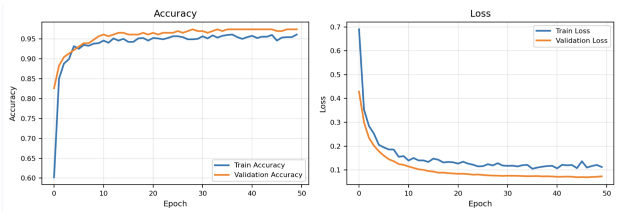

Fig. 3. Training Accuracy and Loss

TABLE VII
EXPERIMENTAL RESULTS

Experiment	Accuracy	Precision	Recall	F1
Exp-1				
Exp-2				
Exp-3				
Exp-4				
Exp-5				
Exp-6				

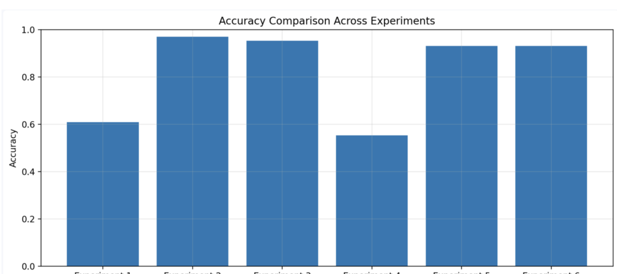

Fig. 4. Comparison of Experimental Accuracy

TABLE VIII
CROSS VALIDATION RESULTS

Scenario	Mean Accuracy	Std Dev
CV-1		
CV-2		
CV-3		
CV-4		
CV-5		
CV-6		

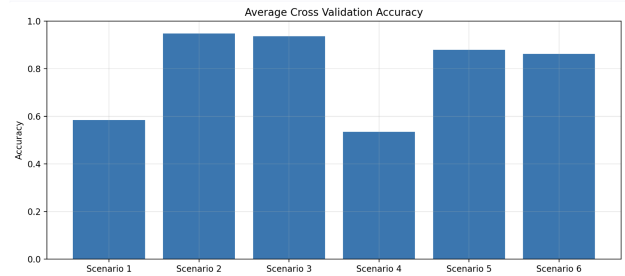

Fig. 5. Cross Validation Accuracy Comparison

C. Cross Validation Results

Cross Validation results indicate that the MinMaxScaler scenario achieved better stability and consistency compared to other normalization methods.

D. Confusion Matrix Analysis

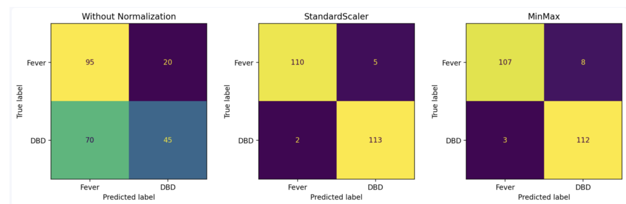

Fig. 6. Confusion Matrix Comparison for Three Normalization Methods

The confusion matrix analysis shows that MinMaxScaler produced fewer classification errors compared to other normalization techniques.

E. ROC Curve Analysis

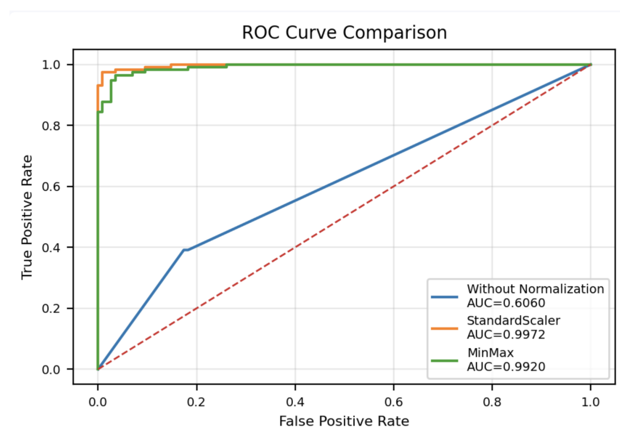

Fig. 7. ROC Curve Comparison for Three Normalization Methods

The ROC Curve results demonstrate that the MinMaxScaler scenario achieved the highest classification capability among all experimental scenarios.

V. CONCLUSION

This study presented a comparative analysis of normalization techniques and cross-validation scenarios for Deep Learning-based Dengue Fever classification. Experimental results demonstrate that normalization significantly affects classification performance. The best performance was achieved using the full dataset with MinMaxScaler normalization. Cross Validation results also indicate good model stability and generalization capability. The proposed Deep Learning model demonstrates promising potential for healthcare decision support systems.

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